Data Engineering and MLOps Assignment – 2

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1. Which run had the lowest latency, and why might that be?

- run_batch1 had the lowest latency of approximately 45 ms (0.045 seconds), according to the MLflow screenshots in the uploaded PDF.
- run_batch4 took approximately 26 ms (0.026 seconds), and run_batch2 took approximately 26 ms (0.026 seconds). The exact values are difficult to read from the screenshots.
- The runs used different batch sizes, which can affect the time taken to process the sentences.
- Larger batch sizes can reduce the overhead of making separate model calls.
- The first run may also take additional time due to model loading and initialization.

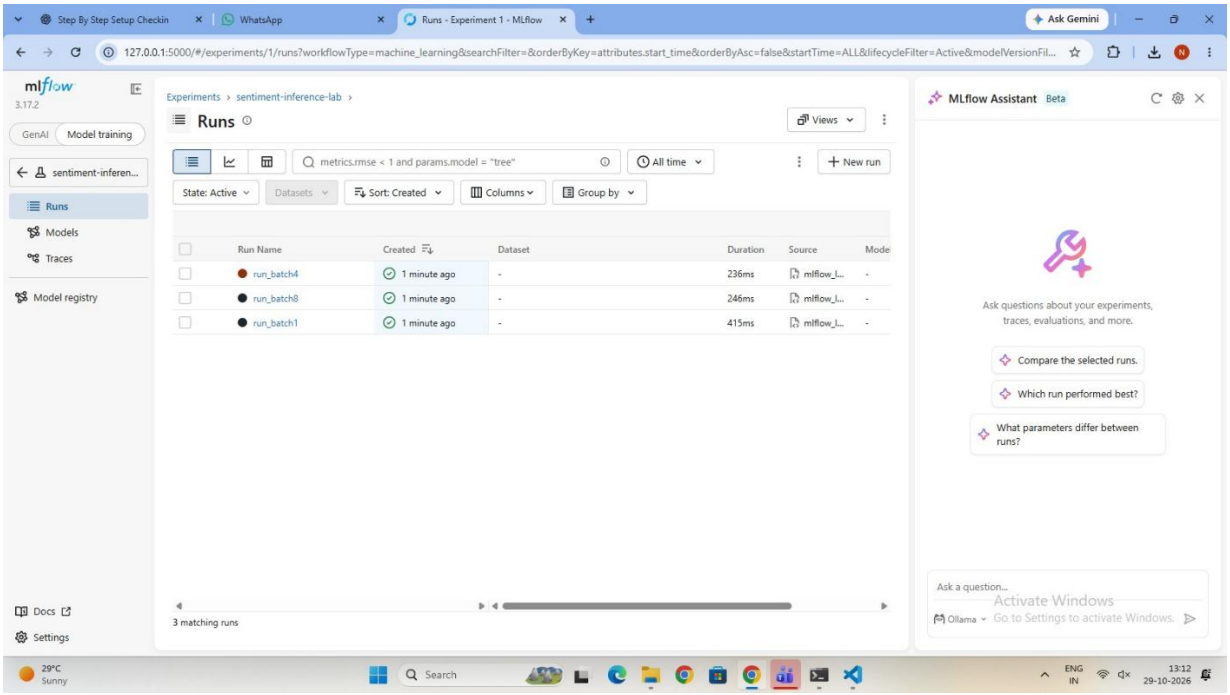
2. Did accuracy change between runs? Why or why not?

- No, accuracy remained at 1.0 (100%) in all three runs.
- All runs used the same pre-trained DistilBERT sentiment analysis model.
- The same dataset of 10 sentences was used for evaluation.
- Changing the batch size affects processing speed but does not necessarily change the model's predictions.
- The average confidence and total latency were logged in MLflow to compare the performance of the runs.

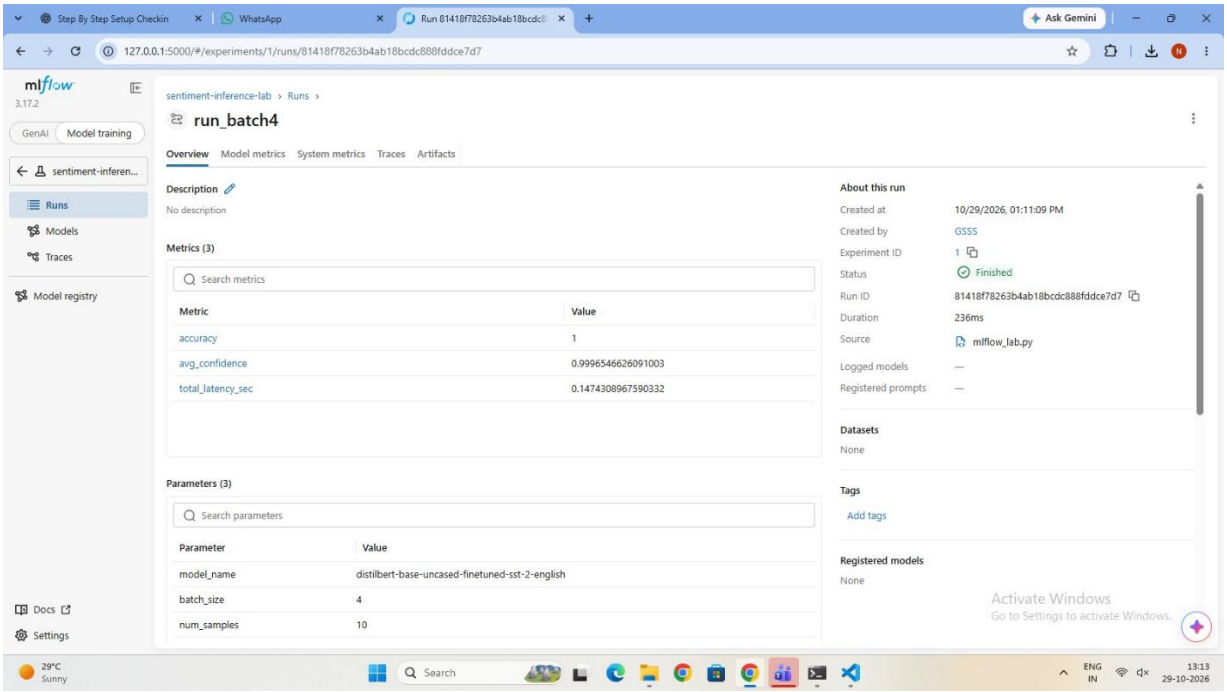
3. Name two benefits of using MLflow here instead of printing results in the terminal.

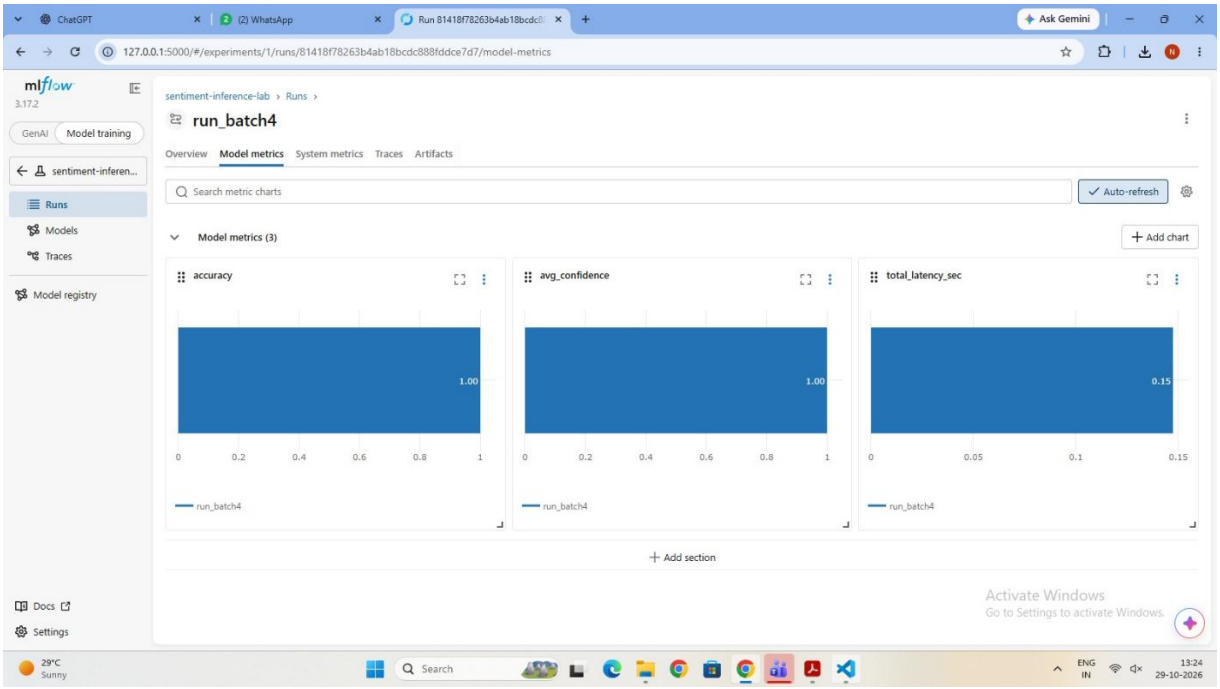
- Experiment tracking and record-keeping: MLflow saves the parameters, metrics such as accuracy and latency, and prediction artifacts (predictions.csv) for each run, making it easier to review the results later.
- Easy comparison and visualization: The MLflow UI allows us to compare multiple runs using tables and charts, helping us understand the differences in model performance without manually examining terminal output.

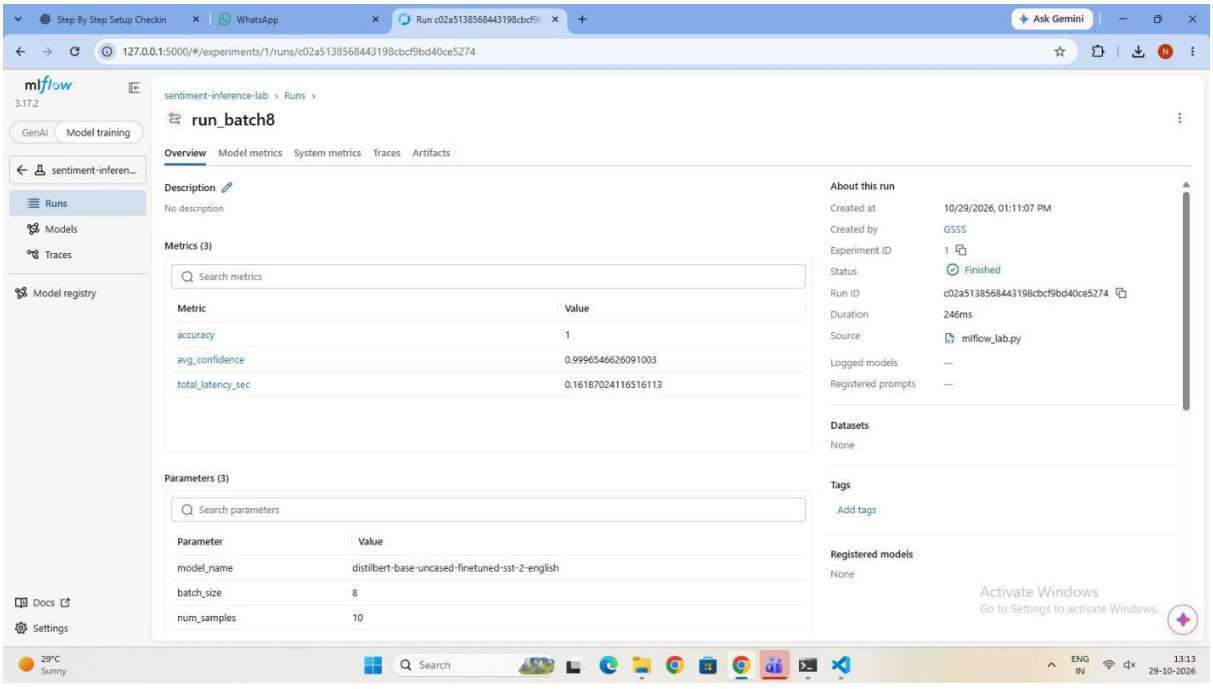
Screenshot of the MLflow runs table:

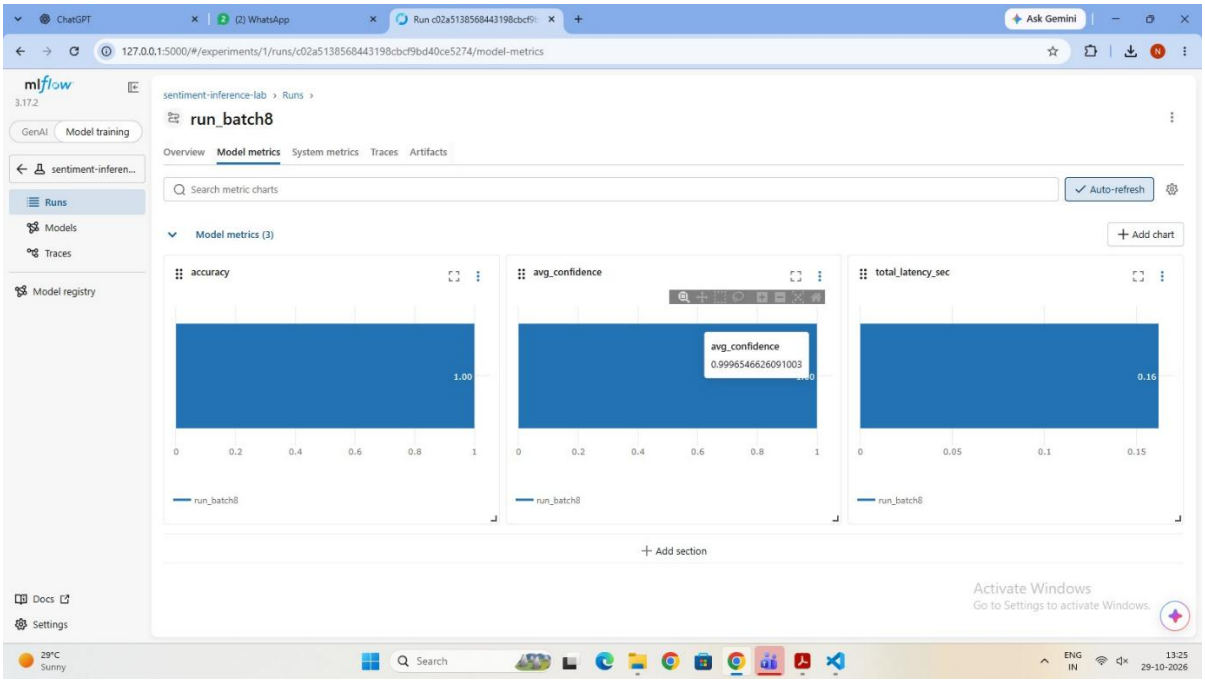


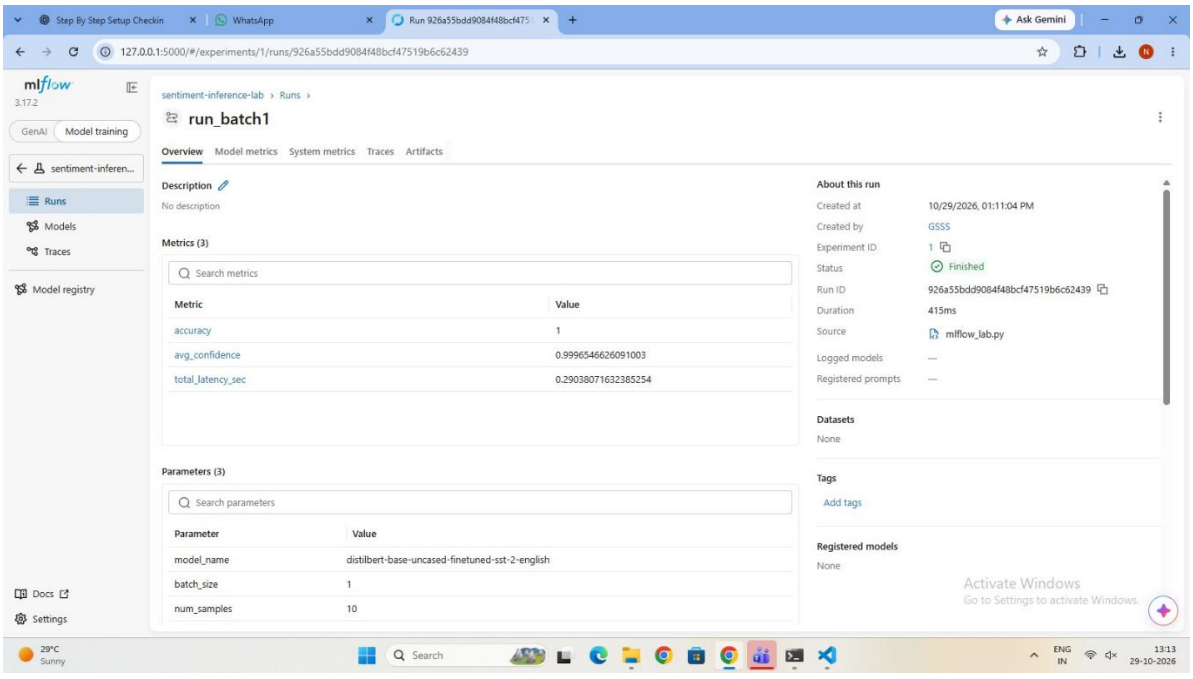
ML Flow Full Screen

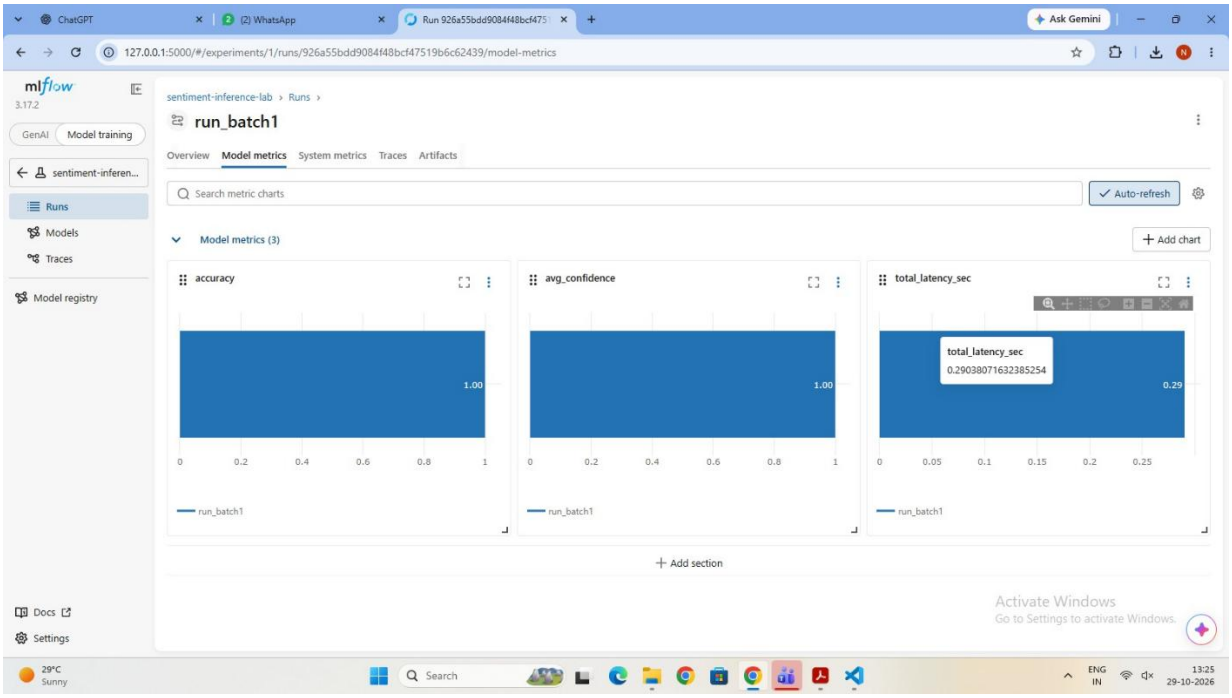



ChatGPT
(2) WhatsApp
Run 81418f78263b4ab18bcdc
Ask Gemini
127.0.0.1:5000/#/experiments/1/runs/81418f78263b4ab18bcdc888fddce7d7/model-metrics
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
sentiment-inference-lab › Runs ›
run_batch4
Overview
Model metrics
System metrics
Traces
Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
avg_confidence
total_latency_sec
1.00
1.00
0.15
run_batch4
Add section
Docs
Settings
Activate Windows
Go to Settings to activate Windows.
29°C
Sunny
Search
ENG
IN
13:24
29-10-2026

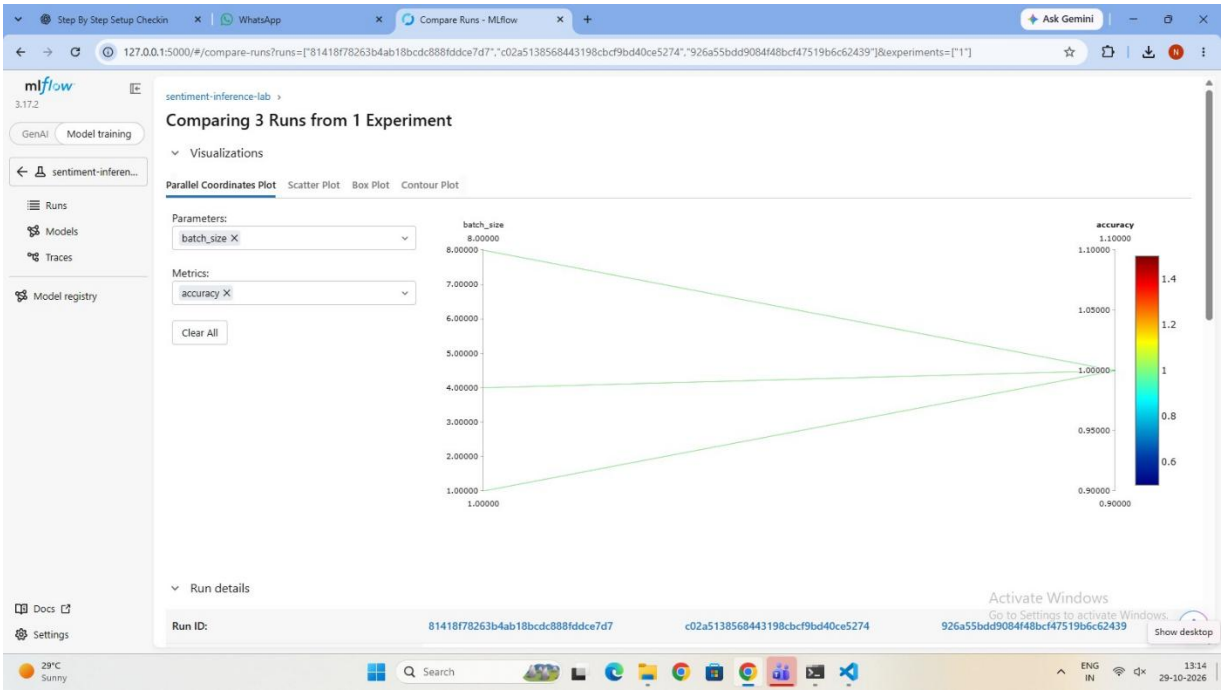

Step By Step Setup Checkin
WhatsApp
Run c02a5138568443198cbcf9
Ask Gemini
127.0.0.1:5000/#/experiments/1/runs/c02a5138568443198cbcf9bd40ce5274
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
sentiment-inference-lab › Runs ›
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Description
No description
Metrics (3)
Search metrics
Metric
Value
accuracy
1
avg_confidence
0.9996546626091003
total_latency_sec
0.16187024116516113
Parameters (3)
Search parameters
Parameter
Value
model_name
distilbert-base-uncased-finetuned-sst-2-english
batch_size
8
num_samples
10
About this run
Created at
10/29/2026, 01:11:07 PM
Created by
GSSS
Experiment ID
1
Status
Finished
Run ID
c02a5138568443198cbcf9bd40ce5274
Duration
246ms
Source
mlflow_lab.py
Logged models
Registered prompts
Datasets
None
Tags
Add tags
Registered models
None
Docs
Settings
Activate Windows
Go to Settings to activate Windows.
29°C
Sunny
Search
ENG
IN
13:13
29-10-2026


ChatGPT
(2) WhatsApp
Run c02a5138568443198cbcf9
Ask Gemini
127.0.0.1:5000/#/experiments/1/runs/c02a5138568443198cbcf9bd40ce5274/model-metrics
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
sentiment-inference-lab › Runs ›
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
avg_confidence
total_latency_sec
avg_confidence
0.9996546626091003
1.00
0.16
run_batch8
Add section
Docs
Settings
Activate Windows
Go to Settings to activate Windows.
29°C
Sunny
Search
ENG
IN
13:25
29-10-2026

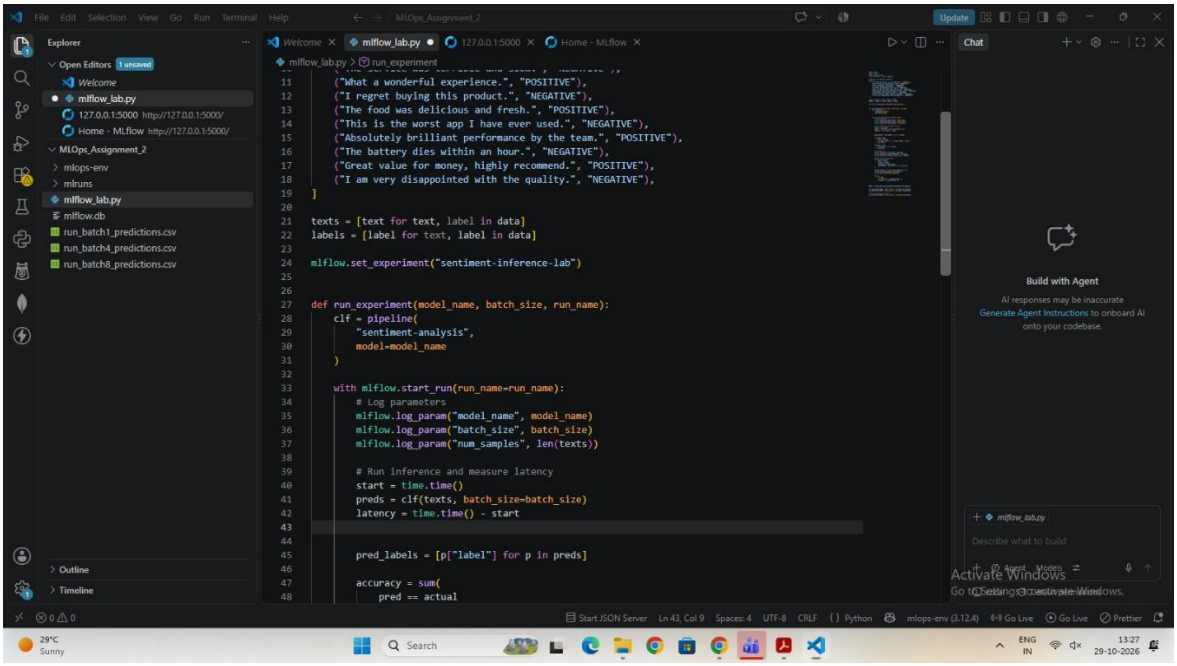

Step By Step Setup Checkin
WhatsApp
Run 926a55bdd9084f48bcf475
Ask Gemini
127.0.0.1:5000/#/experiments/1/runs/926a55bdd9084f48bcf47519b6c62439
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
sentiment-inference-lab › Runs ›
run_batch1
Overview
Model metrics
System metrics
Traces
Artifacts
Description
No description
Metrics (3)
Search metrics
Metric
Value
accuracy
1
avg_confidence
0.9996546626091003
total_latency_sec
0.29038071632385254
Parameters (3)
Search parameters
Parameter
Value
model_name
distilbert-base-uncased-finetuned-sst-2-english
batch_size
1
num_samples
10
About this run
Created at
10/29/2026, 01:11:04 PM
Created by
GSSS
Experiment ID
1
Status
Finished
Run ID
926a55bdd9084f48bcf47519b6c62439
Duration
415ms
Source
mlflow_lab.py
Logged models
—
Registered prompts
—
Datasets
None
Tags
Add tags
Registered models
None
Docs
Settings
Activate Windows
Go to Settings to activate Windows.
29°C
Sunny
Search
ENG
IN
13:13
29-10-2026



Screenshot of the Comparison View:



mlflow_lab.py



```
    ("What a wonderful experience.", "POSITIVE"),
    ("I regret buying this product.", "NEGATIVE"),
    ("The food was delicious and fresh.", "POSITIVE"),
    ("This is the worst app I have ever used.", "NEGATIVE"),
    ("Absolutely brilliant performance by the team.", "POSITIVE"),
    ("The battery dies within an hour.", "NEGATIVE"),
    ("Great value for money, highly recommend.", "POSITIVE"),
    ("I am very disappointed with the quality.", "NEGATIVE"),
]

texts = [text for text, label in data]
labels = [label for text, label in data]

mlflow.set_experiment("sentiment-inference-lab")

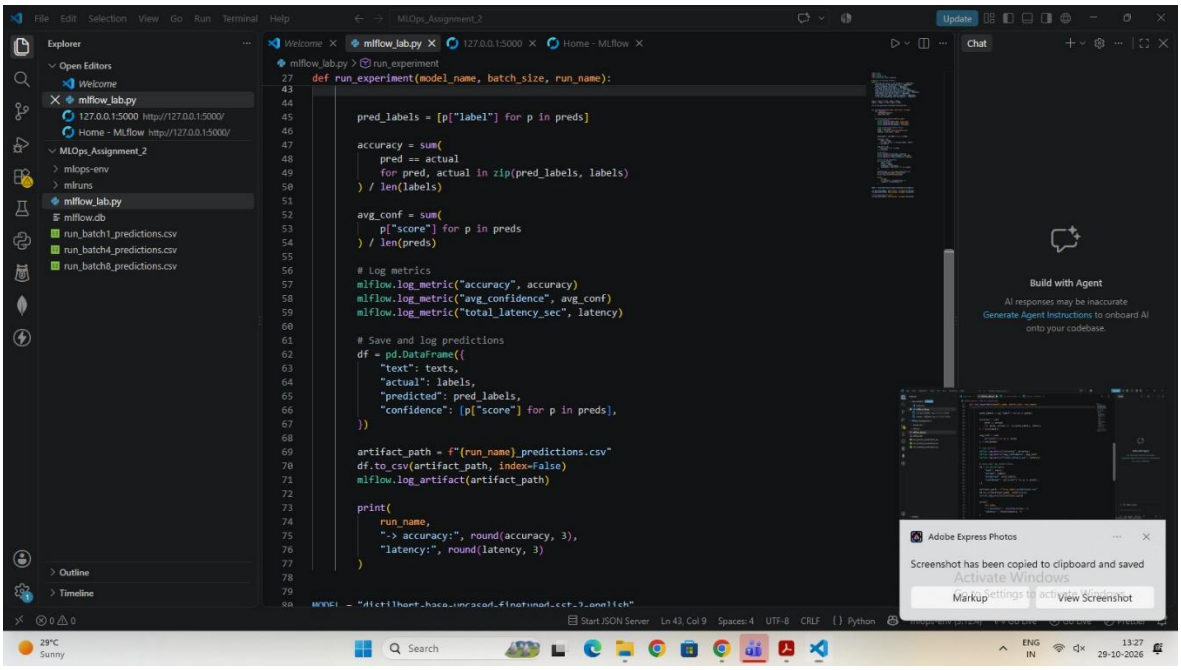

def run_experiment(model_name, batch_size, run_name):
    clf = pipeline(
        "sentiment-analysis",
        model=model_name
    )

    with mlflow.start_run(run_name=run_name):
        # Log parameters
        mlflow.log_param("model_name", model_name)
        mlflow.log_param("batch_size", batch_size)
        mlflow.log_param("num_samples", len(texts))

        # Run inference and measure latency
        start = time.time()
        preds = clf(texts, batch_size=batch_size)
        latency = time.time() - start


        pred_labels = [p["label"] for p in preds]

        accuracy = sum(
            pred == actual
```



```
def run_experiment(model_name, batch_size, run_name):


        pred_labels = [p["label"] for p in preds]

        accuracy = sum(
            pred == actual
            for pred, actual in zip(pred_labels, labels)
        ) / len(labels)

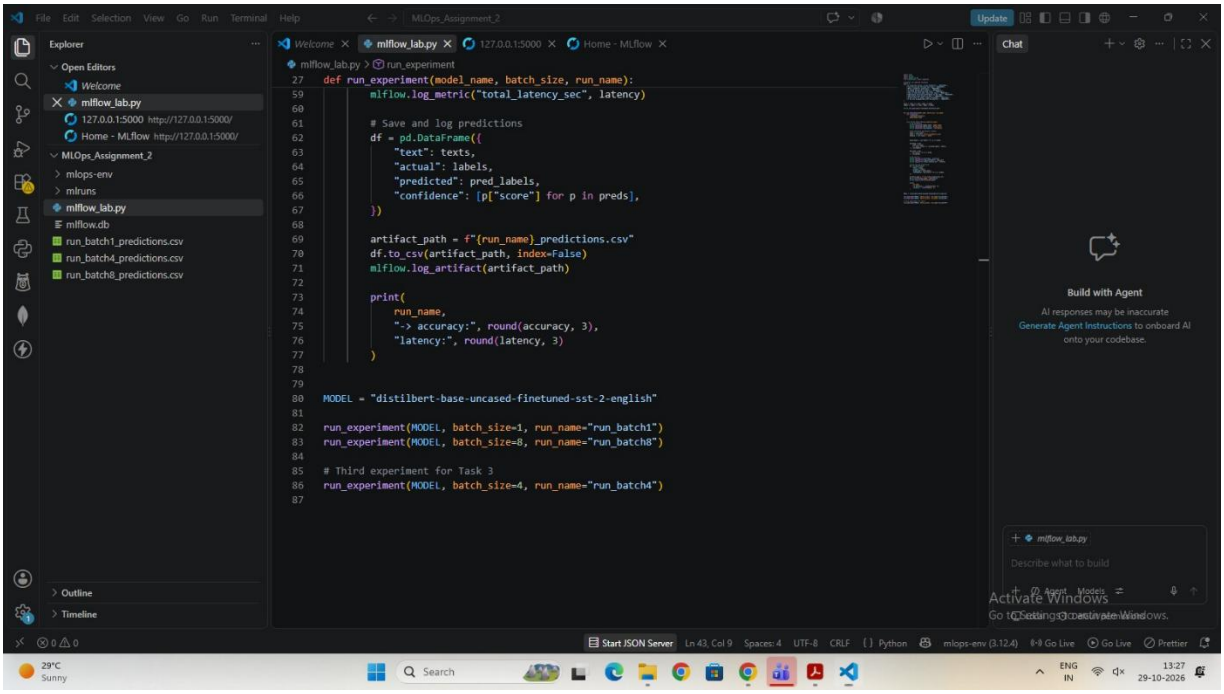
        avg_conf = sum(
            p["score"] for p in preds
        ) / len(preds)

        # Log metrics
        mlflow.log_metric("accuracy", accuracy)
        mlflow.log_metric("avg_confidence", avg_conf)
        mlflow.log_metric("total_latency_sec", latency)

        # Save and log predictions
        df = pd.DataFrame({
            "text": texts,
            "actual": labels,
            "predicted": pred_labels,
            "confidence": [p["score"] for p in preds],
        })

        artifact_path = f"{run_name}_predictions.csv"
        df.to_csv(artifact_path, index=False)
        mlflow.log_artifact(artifact_path)

        print(
            run_name,
            "-> accuracy:", round(accuracy, 3),
            "latency:", round(latency, 3)
        )
```



```python
def run_experiment(model_name, batch_size, run_name):
        mlflow.log_metric("total_latency_sec", latency)

        # Save and log predictions
        df = pd.DataFrame({
            "text": texts,
            "actual": labels,
            "predicted": pred_labels,
            "confidence": [p["score"] for p in preds],
        })

        artifact_path = f"{run_name}_predictions.csv"
        df.to_csv(artifact_path, index=False)
        mlflow.log_artifact(artifact_path)

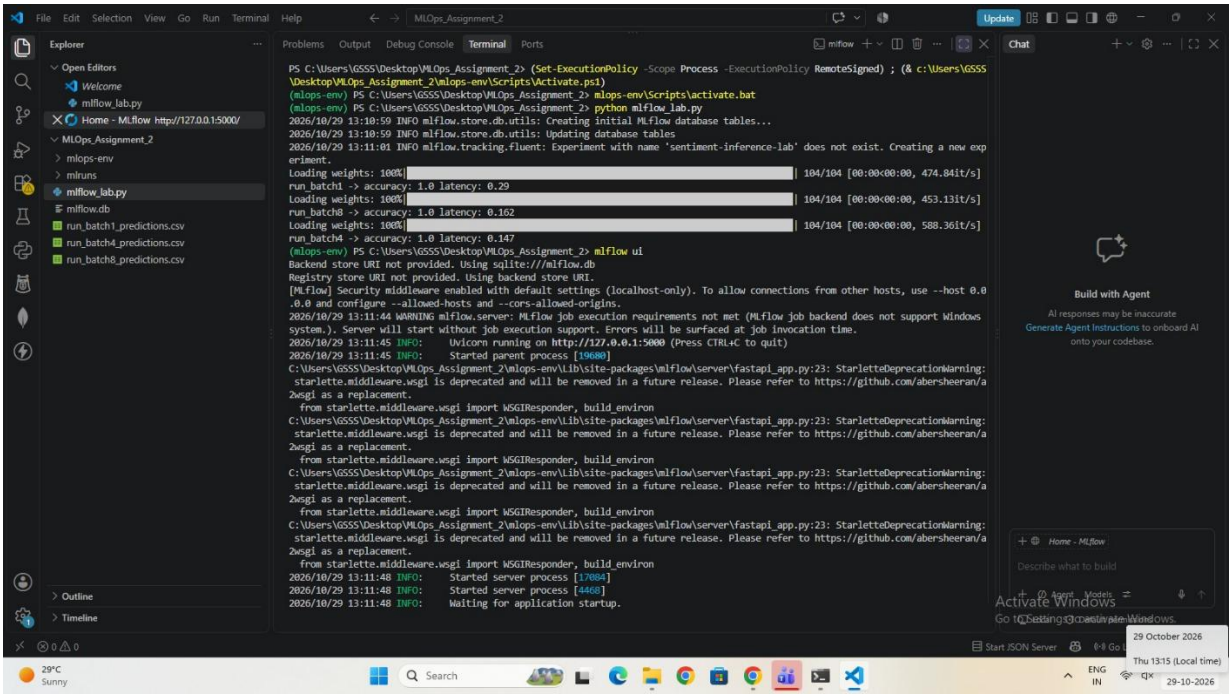
        print(
            run_name,
            "-> accuracy:", round(accuracy, 3),
            "latency:", round(latency, 3)
        )


MODEL = "distilbert-base-uncased-finetuned-sst-2-english"

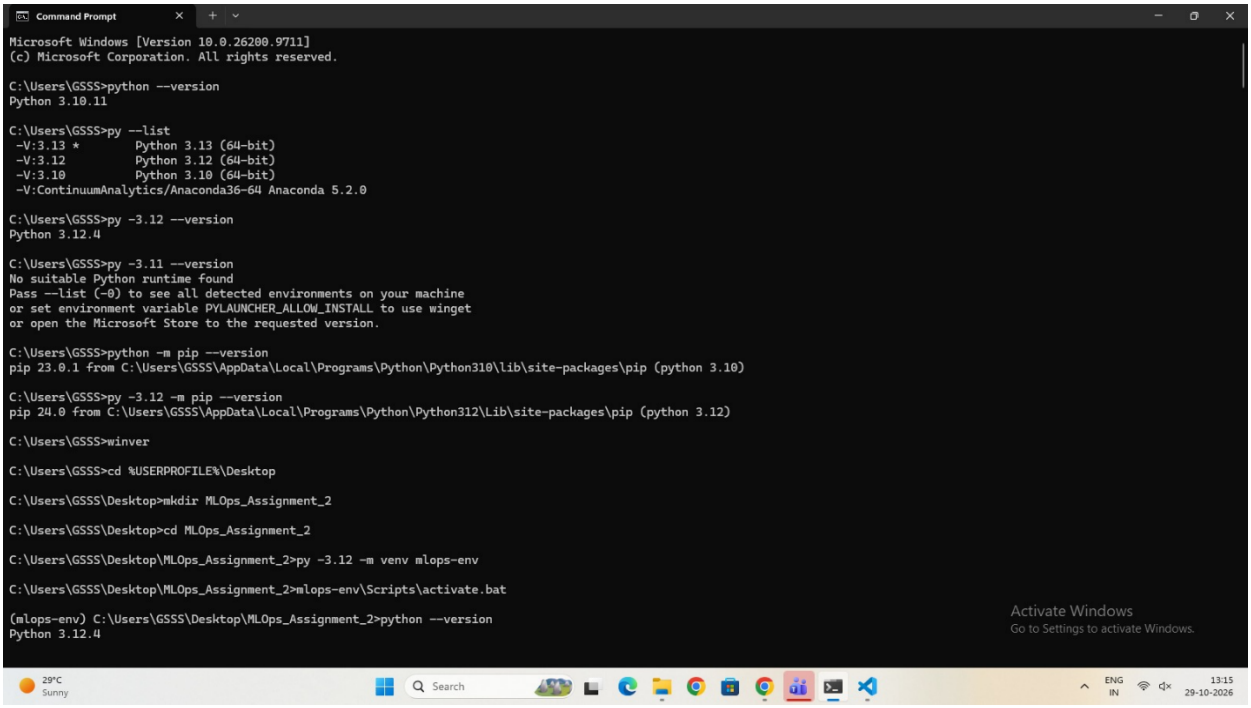
run_experiment(MODEL, batch_size=1, run_name="run_batch1")
run_experiment(MODEL, batch_size=8, run_name="run_batch8")

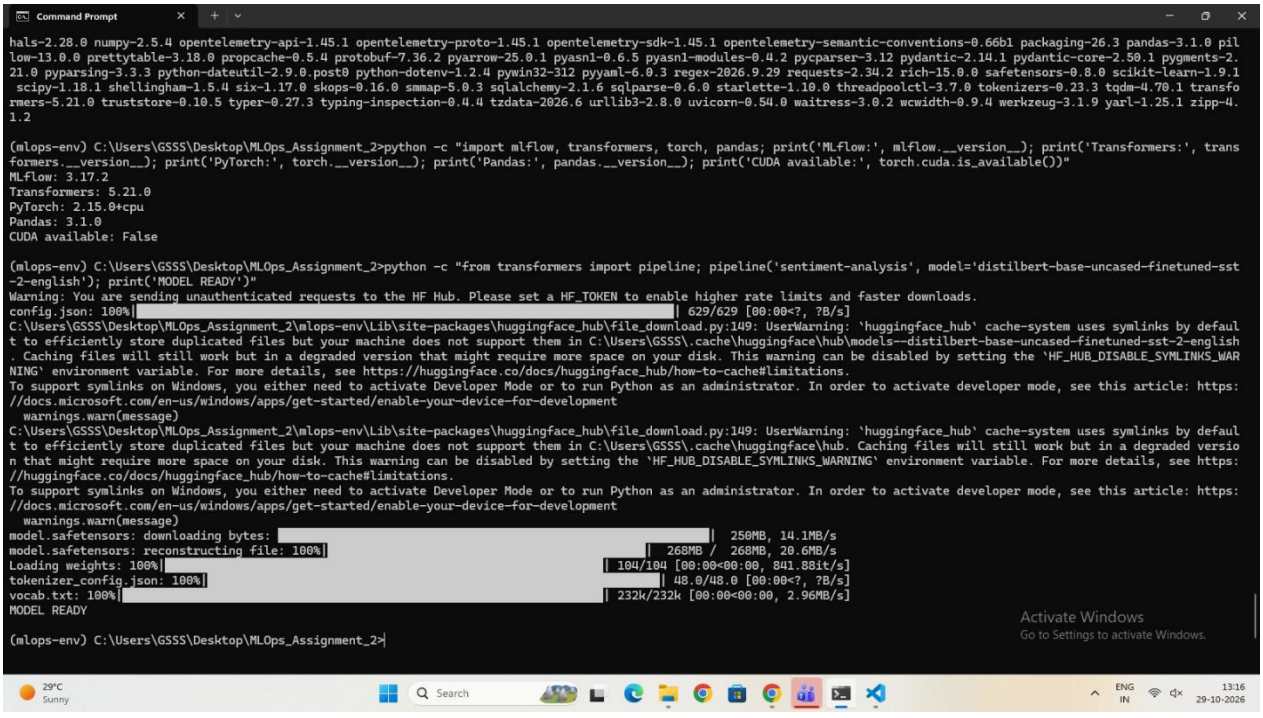
# Third experiment for Task 3
run_experiment(MODEL, batch_size=4, run_name="run_batch4")
```

Execution



```
PS C:\Users\GSSS\Desktop\MLOps_Assignment_2> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& c:\Users\GSSS
\Desktop\MLOps_Assignment_2\mlops-env\Scripts\Activate.ps1)
(mlops-env) PS C:\Users\GSSS\Desktop\MLOps_Assignment_2> mlops-env\Scripts\activate.bat
(mlops-env) PS C:\Users\GSSS\Desktop\MLOps_Assignment_2> python mlflow_lab.py
2026/10/29 13:10:59 INFO mlflow.store.db.utils: Creating initial MLflow database tables...
2026/10/29 13:10:59 INFO mlflow.store.db.utils: Updating database tables
2026/10/29 13:11:01 INFO mlflow.tracking.fluent: Experiment with name 'sentiment-inference-lab' does not exist. Creating a new exp
eriment.
Loading weights: 100%|| 104/104 [00:00<00:00, 474.84it/s]
run_batch1 -> accuracy: 1.0 latency: 0.29
Loading weights: 100%|| 104/104 [00:00<00:00, 453.13it/s]
run_batch8 -> accuracy: 1.0 latency: 0.162
Loading weights: 100%|| 104/104 [00:00<00:00, 588.36it/s]
run_batch4 -> accuracy: 1.0 latency: 0.147
(mlops-env) PS C:\Users\GSSS\Desktop\MLOps_Assignment_2> mlflow ui
Backend store URI not provided. Using sqlite:///mlflow.db
Registry store URI not provided. Using backend store URI.
[MLflow] Security middleware enabled with default settings (localhost-only). To allow connections from other hosts, use --host 0.0
.0.0 and configure --allowed-hosts and --cors-allowed-origins.
2026/10/29 13:11:44 WARNING mlflow.server: MLflow job execution requirements not met (MLflow job backend does not support Windows
system.). Server will start without job execution support. Errors will be surfaced at job invocation time.
2026/10/29 13:11:45 INFO:     Uvicorn running on http://127.0.0.1:5000 (Press CTRL+C to quit)
2026/10/29 13:11:45 INFO:     Started parent process [19680]
C:\Users\GSSS\Desktop\MLOps_Assignment_2\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning:
 starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a
2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\GSSS\Desktop\MLOps_Assignment_2\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning:
 starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a
2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\GSSS\Desktop\MLOps_Assignment_2\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning:
 starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a
2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\GSSS\Desktop\MLOps_Assignment_2\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning:
 starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a
2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
2026/10/29 13:11:48 INFO:     Started server process [17084]
2026/10/29 13:11:48 INFO:     Started server process [4468]
2026/10/29 13:11:48 INFO:     Waiting for application startup.
```


Command Prompt
Microsoft Windows [Version 10.0.26200.9711]
(c) Microsoft Corporation. All rights reserved.
C:\Users\GSSS>python --version
Python 3.10.11
C:\Users\GSSS>py --list
-V:3.13 * Python 3.13 (64-bit)
-V:3.12 Python 3.12 (64-bit)
-V:3.10 Python 3.10 (64-bit)
-V:ContinuumAnalytics/Anaconda36-64 Anaconda 5.2.0
C:\Users\GSSS>py -3.12 --version
Python 3.12.4
C:\Users\GSSS>py -3.11 --version
No suitable Python runtime found
Pass --list (-0) to see all detected environments on your machine
or set environment variable PYLAUNCHER_ALLOW_INSTALL to use winget
or open the Microsoft Store to the requested version.
C:\Users\GSSS>python -m pip --version
pip 23.0.1 from C:\Users\GSSS\AppData\Local\Programs\Python\Python310\lib\site-packages\pip (python 3.10)
C:\Users\GSSS>py -3.12 -m pip --version
pip 24.0 from C:\Users\GSSS\AppData\Local\Programs\Python\Python312\Lib\site-packages\pip (python 3.12)
C:\Users\GSSS>winver
C:\Users\GSSS>cd %USERPROFILE%\Desktop
C:\Users\GSSS\Desktop>mkdir MLOps_Assignment_2
C:\Users\GSSS\Desktop>cd MLOps_Assignment_2
C:\Users\GSSS\Desktop\MLOps_Assignment_2>py -3.12 -m venv mlops-env
C:\Users\GSSS\Desktop\MLOps_Assignment_2>mlops-env\Scripts\activate.bat
(mlops-env) C:\Users\GSSS\Desktop\MLOps_Assignment_2>python --version
Python 3.12.4
Activate Windows
Go to Settings to activate Windows.
29°C
Sunny
Search
ENG
IN
13:15
29-10-2026


Command Prompt
hals-2.28.0 numpy-2.5.4 opentelemetry-api-1.45.1 opentelemetry-proto-1.45.1 opentelemetry-sdk-1.45.1 opentelemetry-semantic-conventions-0.66b1 packaging-26.3 pandas-3.1.0 pillow-13.0.0 prettytable-3.18.0 propcache-0.5.4 protobuf-7.36.2 pyarrow-25.0.1 pyasn1-0.6.5 pyasn1-modules-0.4.2 pycparser-3.12 pydantic-2.14.1 pydantic-core-2.50.1 pygments-2.21.0 pyparsing-3.3.3 python-dateutil-2.9.0.post0 python-dotenv-1.2.4 pywin32-312 pyyaml-6.0.3 regex-2026.9.29 requests-2.34.2 rich-15.0.0 safetensors-0.8.0 scikit-learn-1.9.1 scipy-1.18.1 shellingham-1.5.4 six-1.17.0 skops-0.16.0 smmap-5.0.3 sqlalchemy-2.1.6 sqlparse-0.6.0 starlette-1.10.0 threadpoolctl-3.7.0 tokenizers-0.23.3 tqdm-4.70.1 transformers-5.21.0 truststore-0.10.5 typer-0.27.3 typing-inspection-0.4.4 tzdata-2026.6 urllib3-2.8.0 uvicorn-0.54.0 waitress-3.0.2 wcwidth-0.9.4 werkzeug-3.1.9 yarl-1.25.1 zipp-4.1.2
(mlops-env) C:\Users\GSSS\Desktop\MLOps_Assignment_2>python -c "import mlflow, transformers, torch, pandas; print('MLflow:', mlflow.__version__); print('Transformers:', transformers.__version__); print('PyTorch:', torch.__version__); print('Pandas:', pandas.__version__); print('CUDA available:', torch.cuda.is_available())"
MLflow: 3.17.2
Transformers: 5.21.0
PyTorch: 2.15.0+cpu
Pandas: 3.1.0
CUDA available: False
(mlops-env) C:\Users\GSSS\Desktop\MLOps_Assignment_2>python -c "from transformers import pipeline; pipeline('sentiment-analysis', model='distilbert-base-uncased-finetuned-sst-2-english'); print('MODEL READY')"
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
config.json: 100%| 629/629 [00:00<?, ?B/s]
C:\Users\GSSS\Desktop\MLOps_Assignment_2\mlops-env\Lib\site-packages\huggingface_hub\file_download.py:149: UserWarning: `huggingface_hub` cache-system uses symlinks by default to efficiently store duplicated files but your machine does not support them in C:\Users\GSSS\.cache\huggingface\hub\models--distilbert-base-uncased-finetuned-sst-2-english. Caching files will still work but in a degraded version that might require more space on your disk. This warning can be disabled by setting the `HF_HUB_DISABLE_SYMLINKS_WARNING` environment variable. For more details, see https://huggingface.co/docs/huggingface_hub/how-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate developer mode, see this article: https://docs.microsoft.com/en-us/windows/apps/get-started/enable-your-device-for-development
warnings.warn(message)
C:\Users\GSSS\Desktop\MLOps_Assignment_2\mlops-env\Lib\site-packages\huggingface_hub\file_download.py:149: UserWarning: `huggingface_hub` cache-system uses symlinks by default to efficiently store duplicated files but your machine does not support them in C:\Users\GSSS\.cache\huggingface\hub. Caching files will still work but in a degraded version that might require more space on your disk. This warning can be disabled by setting the `HF_HUB_DISABLE_SYMLINKS_WARNING` environment variable. For more details, see https://huggingface.co/docs/huggingface_hub/how-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate developer mode, see this article: https://docs.microsoft.com/en-us/windows/apps/get-started/enable-your-device-for-development
warnings.warn(message)
model.safetensors: downloading bytes: 250MB, 14.1MB/s
model.safetensors: reconstructing file: 100%| 268MB / 268MB, 20.6MB/s
Loading weights: 100%| 104/104 [00:00<00:00, 841.88it/s]
tokenizer_config.json: 100%| 48.0/48.0 [00:00<?, ?B/s]
vocab.txt: 100%| 232k/232k [00:00<00:00, 2.96MB/s]
MODEL READY
(mlops-env) C:\Users\GSSS\Desktop\MLOps_Assignment_2>
Activate Windows
Go to Settings to activate Windows.
29°C
Sunny
Search
ENG
IN
13:16
29-10-2026